Two-Way Digital Paging System Using Software Defined Radios

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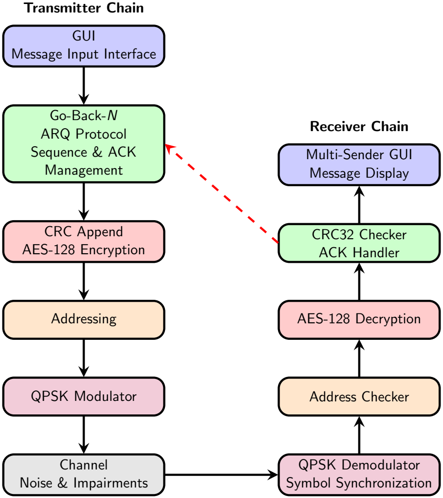
Presentation Structure

Introduction

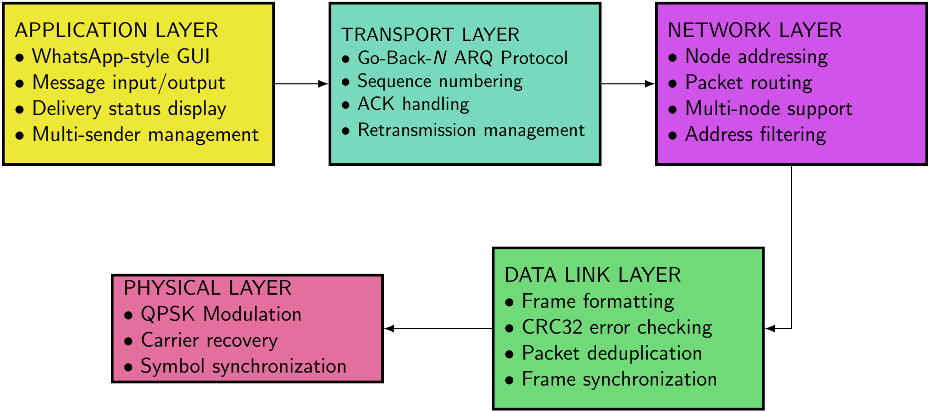
Abstraction

This project is a comprehensive **Software-Defined Radio (SDR)** platform that implements a complete digital communication protocol stack. The system integrates **QPSK physical layer transmission**, **Go-Back-N ARQ** for reliable data transfer, and **AES-128 encryption** for secure communications through intuitive user interfaces. By combining advanced signal processing algorithms with modern protocol design, this project demonstrates a **full-stack approach** that spans from physical layer processing to application-level messaging, making sophisticated telecommunications concepts accessible and operable in real-world scenarios.

Block Diagram



TCP/IP Protocol Stack - 5 Layer Model



Graphical User Interface (GUI)

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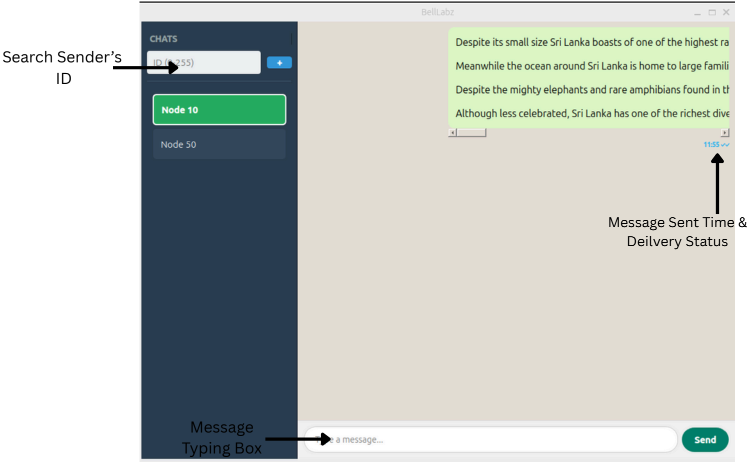


Graphical User Interface (GUI) (Contd...)

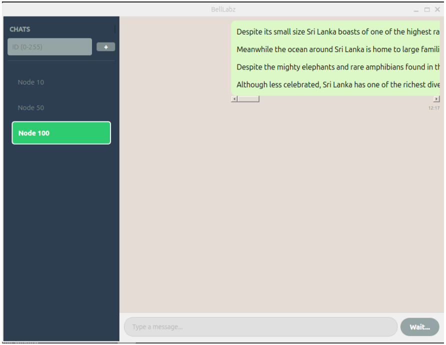


Figure: When Transmitting a Message...

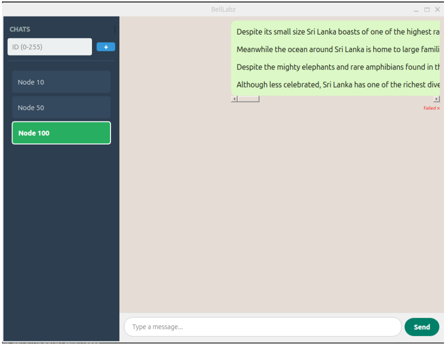


Figure: When transmission fails, sender knows transmission has been failed

Receiver's GUI

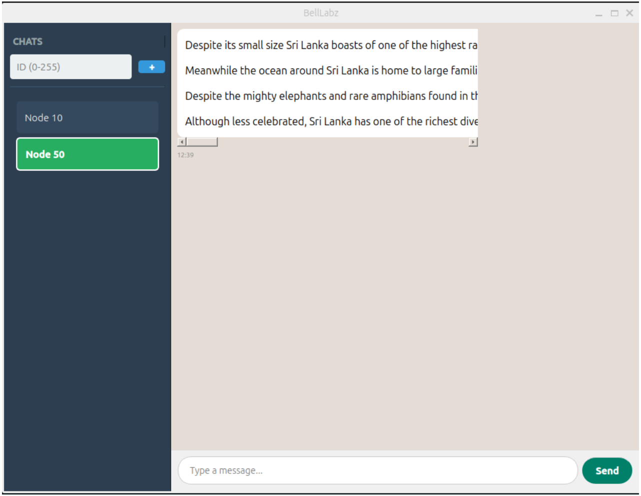
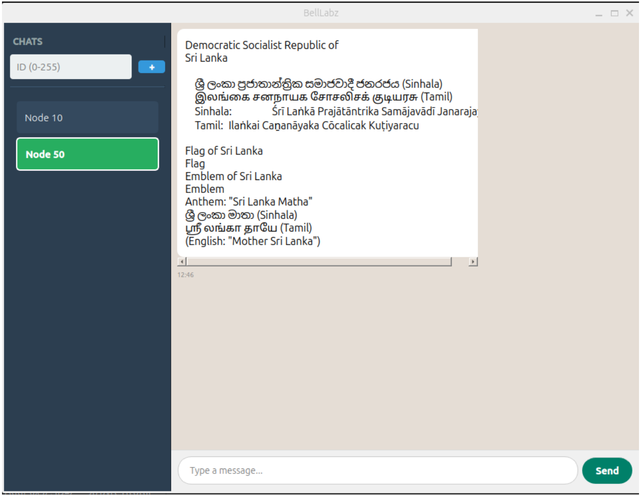


Figure: Receiver's GUI

Multilingual Support with UTF-8 Encoding



Our system breaks language barriers while maintaining full protocol compatibility.

Methodology

Frame Structures

AES 128 Encrypted

Access Code (4 B)	Packet Length (2 B)	Receiver's Address (1 B)	Sender's Address (1 B)	Sq. ID (2 B)	Payload ($\leq N$ B)	CRC- 32 (4 B)

Figure: Message Frame

Access Code (4 B)	Packet Length (2 B)	Sender's Address (1 B)	0xAA	Sq. ID (2 B)	CRC- 32 (4 B)

Figure: Acknowledgement Frame

- The payload length N, can be changed.
- At the end of a message, transmitter transmits a '0' bit stream, then the receiver knows the ending of the message.

ARQ Mechanism

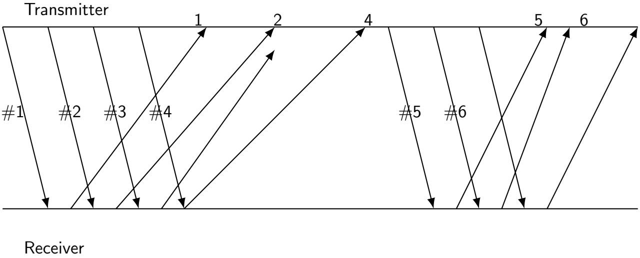
We are using **Go-Back-N ARQ**.

Step 1: Sender can transmit up to N unacknowledged frames.

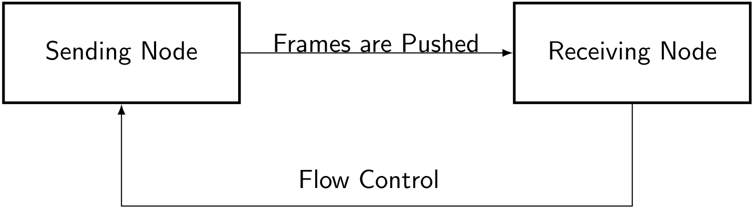
Step 2: Receiver accepts frames in order; if an error occurs, it discards subsequent frames.

Step 3: On timeout or error detection, sender retransmits from the erroneous frame.

Step 4: Process continues until all frames are acknowledged.



Flow Controlling

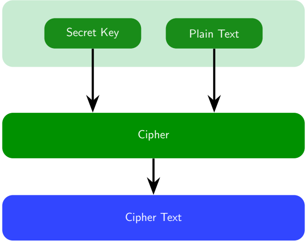


Feedback Mechanism

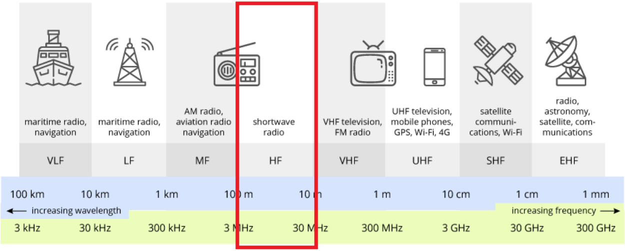
- The data-link layer at the sending node tries to push frames toward the data-link layer at the receiving node.
- Continuous feedback enables real-time flow adjustment.

Dynamic Encryption

- On the transmitter side, depending on the receivers address the AES-128, key is calculated through a special algorithm.
- Each receiver uses a unique decryption key that's specific to it's own address.
- Since other nodes have different keys those nodes won't be able to decrypt it correctly.
- Therefore, other nodes will discard the message.

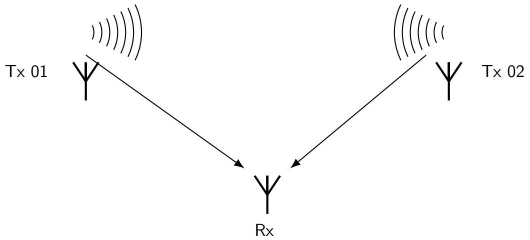


Channelization Using FDMA for Data path/ ACK path



- We used 2.4 & 5.8 GHz ISM bands for the upstream and the downstream respectively.
- Therefore nodes can send data while receiving ACKs.

Multiple Users Handling



- Each node has a unique ID number.
- Receiver keeps a dictionary of buffers, one for each sender.
- Each ACK is sent back to the correct sender.
- Our System Supports transmit messages among up-to 256 (2^8) stations.

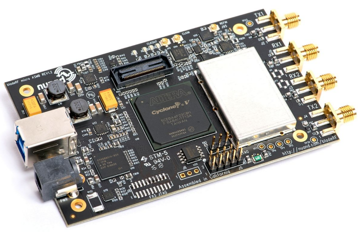
Collision Management

Detection, Notification, and Recovery

1. **Collision Event**
   - Multiple stations transmit simultaneously.
   - Random frame overlap in shared channel.
2. **Failure Detection**
   - CRC validation fails $\rightarrow$ Corrupted data.
   - ACK timeout $\rightarrow$ No confirmation received.
   - Re-try limit exceed $\rightarrow$ Transmitter knows transmission is unsuccessful.
3. **User Awareness**
   - GUI displays **"Failed"** status.
   - Immediate visual feedback.
4. **Recovery Action**
   - Manual retry capability available.

Hardware Testing

Hardware Configuration: Dual bladeRF 2.0 Setup

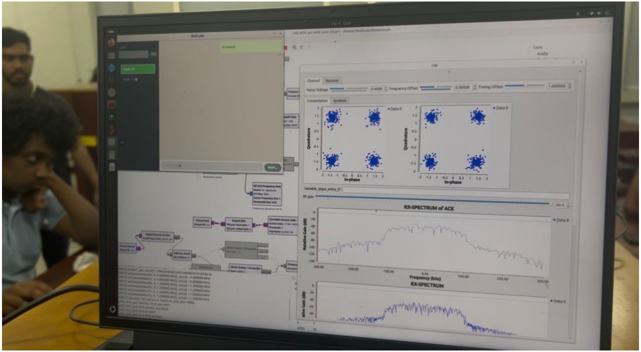


- **Data Transmission Channel:** 2.4 GHz ISM Band
- **Acknowledgment Path:** 5.8 GHz ISM Band

Performance Benefits

- Data and ACK channels cannot collide.
- Simultaneous transmission and acknowledgment.
- Immediate ACK processing without channel contention.
- ACK channel always available for feedback.

System Demonstration



Click to Play Video Demonstration

Main Challenges & Solutions

1. **Hardware Driver Issues**
   - Nuand bladeRF 2.0 drivers unstable on Windows.
   - Driver conflicts with other USB devices.
     Solution: Dual-boot configuration: Windows + Ubuntu.
2. **Synchronization Issues**
   - Small number of bits are transmitted due to the small length of messages.
   - Not enough bits for the constellation diagram to synchronize.
   - Nuand bladeRF 2.0 are best optimized for streaming long files,images,videos etc.
     Solution: Added a very long preamble with random bits so that it's long enough to synchronize

Main Challenges & Solutions (Contd...)

- On the transmitter side, synchronizations bits ($\approx$10 000) are sent along with the entire window.
- According to the Go-Back-N-ARQ algorithm, when transmitting after a timeout period, again synchronizations bits are sent along with the entire window.
- On the receiver side, since the ACK packets are small it sends 4 packets (or lesser number of packets if a certain time passes) along with synchronization bits.

Consequences:

- Data rate decreased significantly.
- Software complexity increased (Due to generating random numbers etc.).

Main Challenges & Solutions (Contd...)

3. Antenna Issues

- Both Wi-Fi antennas and Telescopic Dipole antennas were tested to see which performs best
  Solution: For the frequency used, Wi-Fi antennas performed best.
- But still reliability was low, so the protocols were changed to increase reliability.

Figure: Wi-Fi Antenna

Figure: Telescopic Dipole Antenna

Main Challenges & Solutions (Contd...)

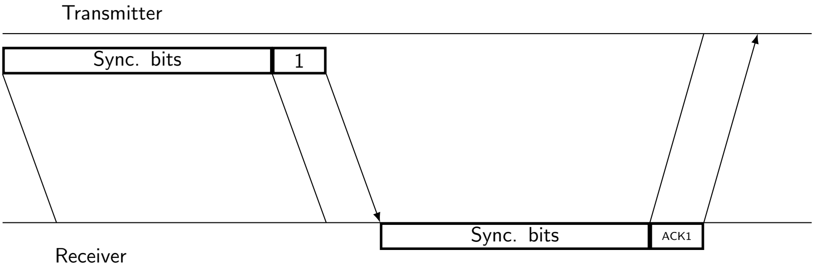


Figure: Optimized ARQ Mechanism

4. Availability and Unpredictability of Nuand BladeRF

- Since only 3 devices were available, it was hard to have access to 2 devices per group to test and debug.
- Unpredictability of the bladeRF devices made it hard to identify errors.

Solution: Half-duplex mode using Ettus Research USRP N210.

Figure: Ettus Research USRP N210

Main Challenges & Solutions (Contd...)

1. Advantages
   - Very reliable for continuous high-rate streaming.
   - Synchronization speed for QPSK modulation is relatively high.
   - Due to buffer mechanism number of bits needed for synchronization is relatively low.
   - Message transmission and acknowledgment reception is guaranteed.
2. Disadvantages
   - Only Half-Duplex mode could be realized with our design.(Because it only has two antennas, but for our design 4 antennas are required)
   - Due to the buffer mechanism even after the transmitter has received the acknowledgment, the messages already buffered at the receivers end result in decreased throughput.(While bladeRF provides real time transmission, USRP causes delays due to buffering mechanism)
   - If we decrease the number of synchronization bits it will solve the buffering problem but then it's hard to synchronize.

Performance Figures

For Nuand BladeRF 2.0

Test Conditions

- **Antenna Type:** Wi-Fi antennas
- **Antenna Gain (TX):** 50 dB
- **Antenna Gain (RX):** 50 dB
- **Test Distance:** 1 m
- **Modulation:** QPSK
- **SPS:** 4

Configuration Constraints

- **Sample Rate:** 1.2 MHz
- **Modulation Rate:** 300 kSymbols/s
- **Retry Limit:** 100
- **Synchronization Bits** 10,000 bits
- **Security:** AES-128 Encryption

Performance Metric	Value	Units
Data Rate	175	Bps
Frame Success Rate	63.95	%
ACK Success Rate	42.73	%
CRC Failure Rate	36.05	%
Maximum Distance	10 (?)	m

Conclusion

Future Works

- **Multimedia Support:** Video, image, PDF and voice transmission
- **Carrier Sense:** Using CSMA/CA for avoiding collisions.
- **Security:** Athentication protocols
- **Data Rate:** Finding better ways of synchronization using lesser number of bits
- **Priority-based message handling:** Based on weather its an emergency give messages priority
- **Multiplexing:** Code Division Multiplexing (CDMA) for simultaneous transmission between multiple users without collisions

Goal: Enterprise-grade wireless communication platform

References

- J.-M. Friedt and H. Boeglen, *"Communication Systems Engineering with GNU Radio: A Hands-on Approach"*. Hoboken, NJ, USA: John Wiley & Sons, 2025.
- GNU Radio Project, "Tutorials," *GNU Radio Documentation*. Available: https://wiki.gnuradio.org/index.php/Tutorials. Accessed: Nov. 10, 2025.

Thank You!